

Project plan

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MAT 421

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InstaCart Data Analysis

Overview

A business from the United States called Instacart provides grocery delivery and pickup services in both the United States and Canada [4]. Instacart, a grocery delivery and shopping app, aims to make it simple to keep your fridge and cabinets stocked with your favorite foods and necessities whenever you need them [2]. You can use the Instacart app to place your order, and personal shoppers will review it, shop for you in-store, and deliver your items. The data set for this project consists of a relational collection of files characterizing the orders of consumers over time [1]. The problem at hand is that the InstaCart company wants to predict the number of reorders that will be within the next period. The data given to us is a collection of files which contain random variables that aren't connected to each other [5]. The data given can be used to predict and investigate several significant aspects that have not yet been investigated [1]. We want to identify potential trends, predict numerical data, and interpret existing data using numerical procedures and computation throughout python.

Existing Research

Each entity (customer, product, order, aisle, etc.) has an associated unique id within the given data set concerning instacart [1]. The highest performing and lowest performing products are highlighted with the instacart data set and information found on the web [5]. The coronavirus global epidemic caused a 3-fold boost in revenue from 2019 to 2020, but InstaCart's overall revenue is anticipated to grow much more slowly in the US going forward [2]. The grocery delivery company brought in almost 26 billion dollars in revenue in 2021. The company's domestic market sales could total 40.5 billion dollars by 2024 [2]. It is crucial to keep in mind that no such analyses have been done to forecast any potential future models. The company's tremendous 12-month growth it had early in the pandemic is no longer able to be sustained [5]. Since the second quarter of 2021, Instacart's sales have consistently lagged below those of the same period the previous year. Sales for Instacart decreased by 4% in the first quarter of 2022 compared to the same period in 2021 [5]. The reality that customers are purchasing less at the supermarket delivery service than they were at the start of the epidemic is another probable contributing factor to Instacart's falling sales growth [4]. Locally, Instacart is still experiencing sales growth in a few metropolitan areas. Dallas, Philadelphia, New York, as well as Houston were the four largest metropolitan areas by population with positive 12-month growth rates in Instacart sales [3]. No such predictions on the future of the instacart sales have been made. The predictions can be made by using numerical analysis in python as well as other mathematical functions that are within python that will help us create strong prediction models and other important information that we want to extract within the data set.

Problems

1. We want to predict the future sales for instacart given the data set available online [1].

2. We want to modify the data in order to better understand the relationship between the numerical variables so we can attempt to predict the data through graphical representations.
3. Using linear equations and regression we want to predict the future reorder quantities for the commonly specified products.
4. Using differentiation and optimization we want to assess the current existing models of the instacart datasets and tackle the trouble that exists within them.

Models and Numerical Methods

I. Regression

Finding the variables that have an effect on an interest issue can be done with accuracy using regression analysis. The analysis of the Instacart data set will be aided by this. Regression analysis will enable us to confidently ascertain which elements are most important, which factors can be disregarded, and how the various factors interact [6].

II. Linear Equations

The analysis of the Instacart patterns and predictions will heavily rely on linear equations. They will describe the links between variables in the data, form hypotheses, determine rates, and perform transformations [6].

III. Differentiation & Optimization

We can make sense of many patterns and numerical values in the data by differentiating them. The goal of optimization is to provide the optimal model in light of a list of prioritized requirements or restrictions. This will help us with our forecasting models [6].

IV. Root Finding

Rooting finding will aid us in understanding the key boundaries of the data as well as form specific conclusions on products, reorders, etc [6].

Expectations

- I. Understand the key relationships between the numerical variables across the data set.
- II. Form definitive conclusions on the data set using created models.
- III. Predict the future revenue by instacart using the existing data and numerical methods.
- IV. Remodify the data so it is presented in a user friendly manner that can be used for later analysis if needed.

Works Cited

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